

SOP-504: STATUTORY INTEGRITY STANDARDS FOR REFINERY PRESSURE VESSELS

Standard Code: NRC-HSE-SOP-504-REV-04 | Classification: STRICTLY CONFIDENTIAL

SECTION 1.0 — OBJECTIVE & JURISDICTION

This Standard Operating Procedure (SOP) establishes mandatory safety envelopes, minimum retirement wall thicknesses, and shutdown criteria for all high-temperature pressure vessels, columns, and heat exchangers across refinery operational units.

SECTION 3.2 — MINIMUM RETIREMENT WALL THICKNESS CRITERIA

Under no operational circumstances shall any pressure boundary component operate below the calculated minimum retirement thickness (t_{min}) determined in accordance with ASME Boiler and Pressure Vessel Code Section VIII. Any measurement indicating wall thickness below t_{min} constitutes an IMMEDIATE CRITICAL NON-CONFORMANCE requiring immediate de-pressurization and formal approval by the Refinery Safety Director before re-pressurization.

SECTION 4.1 — PRESSURE RELIEF VALVE (PRV) MANDATORY TOLERANCE

All PRVs protecting hydrocarbon distillation columns must be bench-calibrated every 12 months. Allowable set-pressure deviation is strictly limited to $\pm 1.5\%$. Any premature seat leakage below 90% of set-pressure or reseal delay exceeding 3.0 seconds requires immediate valve replacement with a certified backup unit prior to start-up.

SECTION 5.4 — CLADDING INTEGRITY & NAPHTHENIC CORROSION

Where crude feeds contain Total Acid Number (TAN) exceeding 0.5 mg KOH/g, stainless steel cladding delamination exceeding 100 cm² requires shutdown weld repair. Operating unclad carbon steel at temperatures above 230°C in high-TAN service is strictly prohibited.

Approved By: Dr. A. Sharma, Director of Plant Safety & Oversight | Approved: 2025-06-01